LECTURE 14

11 October 2023 08:34

MVT: If f is cont. on $[a,b]$ & diff. on (a,b) then $\exists\ c \in (a,b)$ s.t. $f'(c) = \dfrac{f(b)-f(a)}{b-a}$

Eg: Verify MVT for $f(x) = x^2 - 3x + 2$ on $[-2,3]$

$f \to$ polynomial $\Rightarrow$ cont. & diff. on $[-2,3]$

Find a $c \in [-2,3)$ s.t. $f'(c) = \dfrac{f(3)-f(-2)}{3+2}$

$2c - 3 = \dfrac{-12}{5} = \dfrac{-10}{5} = -2$

$2c - 3 = -2$

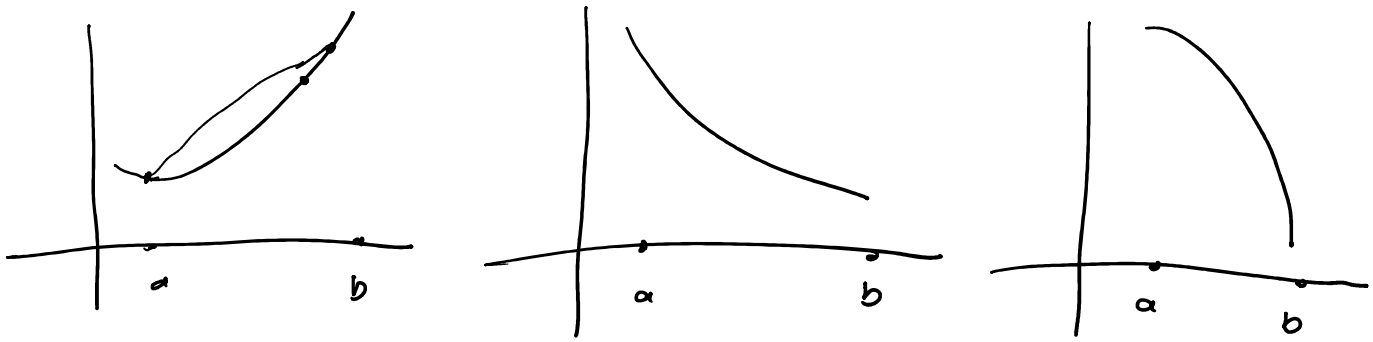
$2c = 1$

$c = 1/2$

Eg: Verify Rolle's thm: $f(x) = x(x-3)^2$ $[0,3]$

$f'(c) = 0$ $\quad c = 1$ & $\cancel{3}$ $\quad c \in (0,3)$

Thm: If f is diff. & $f'(x) = 0\ \forall x \in$ interval $f \equiv$ const.



Convex : chord lies above the curve :

① A diff. f^n. is convex iff $f'(x) \geq 0 \quad \forall x \in [a,b]$

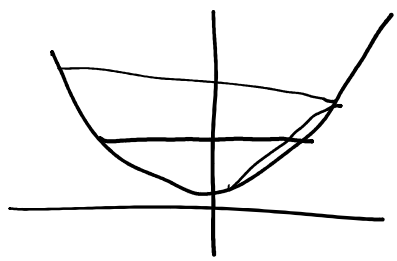
② A two times diff. f^n. is convex iff $f'' \geq 0$

③ If f^n. is ~~conti~~ convex on $(a,b) \Rightarrow f$ is conti on (a,b)

Eg. $f(x) = x^2$

$f'(x) = 2x$

$f''(x) = 2 > 0$



L-Hospital Rule : $\frac{0}{0}$, $\frac{\infty}{\infty}$ form. $\left\{ \begin{matrix} \infty - \infty & 0^0 \\ 1^\infty & \infty^0 \end{matrix} \right\}$

① $\lim_{x \to 0} \frac{1-\cos x}{x^2} \left(\frac{0}{0}\right)$

$= \lim_{x \to 0} \frac{\sin x}{2x} \left(\frac{0}{0}\right)$

$= \lim_{x \to 0} \frac{\cos x}{2} = \frac{1}{2}$

② $\lim_{x \to 0} \frac{\log \sin x}{\log x} \left(\frac{\infty}{\infty}\right)$

$= \lim_{x \to 0} \frac{\frac{\cos x}{\sin x}}{\frac{1}{x}} = \lim_{x \to 0} \frac{x \cos x}{\sin x} \left(\frac{0}{0}\right)$

$= \lim_{x \to 0} \frac{-x \sin x + \cos x}{\cos x} = 1$

Max. min. of $f(x)$:

① critical points. $f'(x) = 0 \qquad x = a, b, \ldots$

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② second derivate test ; $f''(a)$ or $f''(b)$ ≤ 0 max

≥ 0 min

$f(x) = x^2 - 3x + 2$

$f'(x) = 0 \Rightarrow 2x - 3 = 0 \Rightarrow x = 3/2$

$f''(x) = 2 > 0$ at $x = 3/2$ min.